



ALLSEIS-OB4C is the premier choice for E&P companies and service providers in search of cutting-edge subsurface imaging technology. Our nodes are carefully designed to offer exceptional efficiency and environmental sustainability, catering to a wide range of survey needs from shallow waters to depths of 1,000 meters.

Product Highlights

- Standard 3-component 14Hz high sensitivity omnidirectional geophone + 3Hz high sensitivity hydrophone, with an option for a 3Hz AlfSeis® omnidirectional geophone, providing exceptional sensing capabilities for submarine full wave field seismic acquisition;
- 4-channel 24-bit $\Delta\Sigma$ ADC converter with configurable sampling rate and preamplifier gain, offering comparable acquisition parameters to leading international products;
- Built-in high precision thermostatic crystal (OCXO) for timing accuracy within $\pm 2\text{ms}$ over 90 days;
- Built-in test signal generator for on-site system diagnostics, including noise levels, dynamic range, geophone characteristics, etc.;
- Large capacity rechargeable lithium battery supporting 90 days of continuous recording (2,160 hours);
- Multiple data output formats supported, including SEG-D, SEG-Y, Mini-SEED, SAC, etc.;
- Internal electronic tag for efficient asset and inventory management;
- Compact and lightweight design (29D × 14H cm, 13.8kg in air, 7kg in water) for easy transportation and deployment.



General Features

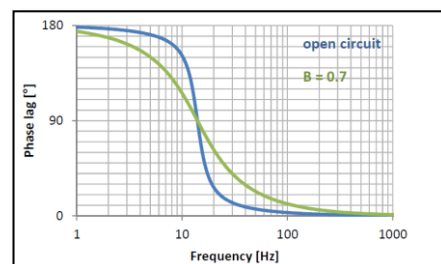
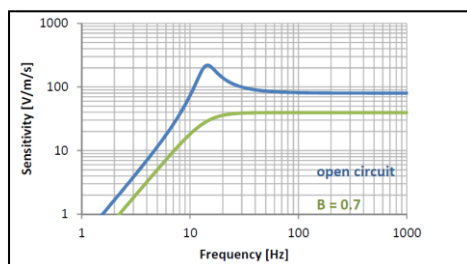
Sensor	3-component 14Hz omnidirectional high sensitivity geophone(Gimbal) Optional 3Hz active force feedback omnidirectional broadband geophone
Hydrophone	3.4Hz broadband high sensitivity hydrophone
Digitalization	4-channel 24-bit Simultaneous-Sampling, Delta-Sigma ADC
Storage capacity	64GB
Battery Life	Internal rechargeable Li-ion battery (750WH) Over 2160hours (90days) continuous recording
Timing Accuracy	+/- 1 μs (continuous GNSS disciplined) +/- 2ms (OCXO within 90 days)
Automated Tests	Internal testing signal generator for function and performance self-inspection
LED Status Indicator	OBN health, GNSS time synchronization status
Data Harvest	Data downloading/battery charging rack
Weight	13.8kg (in air), 7kg (in water)
Size	Diameter: 29cm × Height: 14cm
Operating Temperature	-10°C ~ +45°C
Operating Depth	0~1000m

Acquisition Specifications

ADC Resolution	24-bit
Sampling Interval	0.5ms, 1ms, 2ms, 4ms
Preamplifier Gain	x1, x4, x16 (0dB, 12dB, 24dB)
Gain Accuracy	0.1%
Full Scale Input Signal	$\pm 2.5\text{Vp-p}$ @ x1 gain $\pm 625\text{mVp-p}$ @ x4 gain $\pm 156\text{mVp-p}$ @ x16 gain
Instantaneous DR (1ms, typical value)	125dB @ x1 gain
Full Scale DR	145dB
Equivalent Input Noise (typical value)	1 μV @ x1 gain 0.25 μV @ x4 gain
Harmonic Distortion	-120dB @ 31.5Hz
Frequency Response	0Hz ~ 866Hz @ 0.5ms (-3dB) 0Hz ~ 800Hz @ 0.5ms (-0.1dB)
Anti-alias Filter	Linear filter, 86.6% of Nyquist -3dB bandwidth
Stopband Attenuation	>120dB @Nyquist
Common Mode Rejection	>100 dB
Inter Channel Rejection	>100 dB

14Hz Omni-directional Geophone

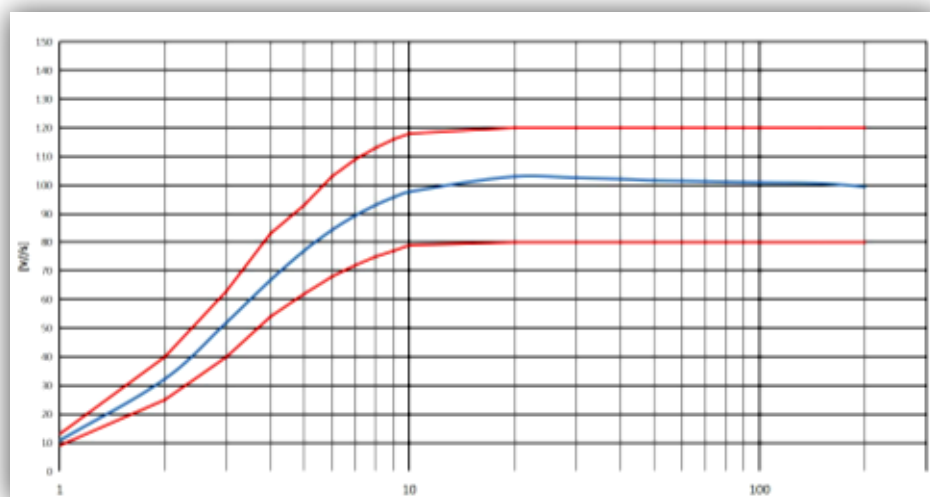
Natural Frequency	14Hz
Sensitivity	80V/m/s
Spurious	> 190Hz
Harmonic Distortion	≤0.1% horizontal, < 0.4% vertical
Coil Resistance	3500Ω
Tilt	0°~360°



14Hz High Sensitivity Omni-directional Geophone Response Curve and Phase Lag

3Hz Omni-directional Wideband Geophone

Natural Frequency	3Hz
Sensitivity	80V/m/s
spurious	> 190Hz
Distortion	< 0.1% horizontal, < 0.4% vertical
Tilt	0°~360°
Equivalent Input Noise	1uVrms@3~190Hz
Full Scale DR	120dB



3Hz Omnidirectional Wideband Geophone Response Curve

High Sensitivity Hydrophone

Frequency Response	3.4Hz@-3dB
Sensitivity	8.9V/bar
Working Depth	≤1000m
Operating Temperature	-10°C~+50°C

Stackable Charging /Downloading Unit

Slots	3
Charging Time	10 hours
Charging Power	350W
Unit Charging Power	100W
Full Download Rate	2GB/min.
Unit Download Rate	0.66GB/min.
Interface	1G Ethernet
Operating Temperature	0°C~+40°C
Size	1U

* Specifications or information subject to change without notice

